

1ª Week: Meeting 22/06/2026



Gabriel de Carvalho 15/06/2026, 09:27

TAREFA

- 1- Escolha dos termos usados na busca
- 2- Busca realizada em quais buscadores (ex. Google Scholar)
- 3- Seleção dos artigos a serem lidos (e quais critérios foram usados)
- 4- Resumo dos selecionados

Como ler um artigo científico:

- 1- Título (Saber se é interessante)
- 2- Resumo/Abstract (Saber se é interessante)
- 3- Conclusão (Entender o que foi descoberto)
- 4- Introdução (Entender melhor o contexto) (editado)

Modelo de Trabalho do Laboratório

Cada aluno é responsável por planejar sua própria semana.

Fluxo semanal:

- 1- O aluno cria uma Issue no GitHub. Na Issue descreve:
 - a- O que pretende realizar;
 - b- Quais entregas pretende produzir.
- 2- O orientador analisa a proposta. A Issue pode ser:
 - a- Aprovada;
 - b- Alterada;
 - c- Rejeitada com sugestões.
- 3- Ao final da semana o aluno registra os resultados obtidos. (editado)

Search performed on the search engine:

Google Scholar

Search term:

Intelligent Adaptive Games

Articles:

Title: Filling the Need for Intelligent, Adaptive Non-Player Characters

Link:

https://www.researchgate.net/profile/Douglas-Maxwell-5/publication/292984684_Filling_the_Need_for_Intelligent_Adaptive_Non-Player_Characters/links/56b3fb6008ae5deb2657e80b/Filling-the-Need-for-Intelligent-Adaptive-Non-Player-Characters.pdf

Title:

To Create Intelligent Adaptive Game Opponent by Using Monte-Carlo for the Game of Pac-Man

Link:

<https://ieeexplore.ieee.org/abstract/document/5366867>

2^a Week: Meeting 29/06/2026

1^o ARTICLE:

Search performed on the search engine:

Google Scholar

Search term:

Intelligent Adaptive Games

Article Title:

To Create Intelligent Adaptive Game Opponent by Using Monte- Carlo for the Game of Pac-Man

Link:

<https://ieeexplore.ieee.org/abstract/document/5366867>

Resume:

Prompt: Summarize this so a 10-year-old can understand it.

Imagine you're playing **Pac-Man**, but instead of the ghosts always following the same rules, they can **think and get smarter**.

Here's what the paper is about in kid-friendly language:

- The researchers wanted to make the ghosts in Pac-Man more fun to play against.
- They used a method called **Monte Carlo**, which is like having the ghosts **imagine lots of possible moves very quickly** before choosing the best one.
- ✂ If the ghosts have **more time to think**, they usually make better choices and become harder to beat. If they have less time, they're easier to beat. This lets the game automatically become easier or harder.
- The problem is that all this thinking takes a lot of computer power and can slow the game down.
- So the researchers trained an **artificial neural network** (a simple kind of AI that learns from examples) by watching the Monte Carlo method play. After learning, the new AI could make good decisions **much faster**.
- In the end, they found that the trained AI played almost as well as the Monte Carlo method but without making the game slow, so it could work better in real games.

In one sentence:

The paper teaches Pac-Man's ghosts to become smarter by practicing lots of pretend games, and then helps them remember what they learned so they can play well without thinking for a long time.

ChatGPT prompt: Summarize the following text in 300 words.

This paper explores how to create a more intelligent and adaptive opponent AI for the game Pac-Man using the Monte Carlo Tree Search (MCT) algorithm. Traditional game opponents often rely on fixed, scripted rules that make their behavior predictable and difficult to adjust. In contrast, Monte Carlo methods enable the AI to evaluate many possible future moves through repeated simulations, allowing it to choose actions that are more likely to succeed. This approach creates opponents that can dynamically adapt their level of difficulty, making gameplay more engaging.

To test this idea, the researchers modified the original Pac-Man game by simplifying some of its rules. Two ghosts cooperate to chase Pac-Man, while Monte Carlo simulations help determine their best movement at each step. The study found that the AI's performance improves as it is given more computation time because additional simulations produce better decisions. However, after approximately 400 milliseconds of computation, further improvements become minimal while causing delays that players would find unacceptable.

Because Monte Carlo simulations require significant computing power, they are not ideal for real-time or online games. To address this limitation, the researchers trained an Artificial Neural Network (ANN) using gameplay data generated by the Monte Carlo algorithm. The neural network learned to imitate the high-quality decisions produced by Monte Carlo while requiring far less computation during gameplay. Inputs to the network included the positions of Pac-Man and the ghosts, distances between them, nearby walls, and other game-state information. The trained model successfully controlled the ghosts with performance comparable to the Monte Carlo approach but with much greater speed and stability.

The authors conclude that combining Monte Carlo simulations with neural network learning provides an effective way to build adaptive game AI. Their approach enables adjustable game difficulty, maintains strong opponent performance, and reduces computational cost, making it suitable for real-time games and potentially applicable to many other game genres beyond Pac-Man.

2° ARTICLE:

Origin:

Reference of article 1

Article Title:

powerGhosts & defensiveGhosts – Enhanced Ghost Team Controller based on Ant Colony Optimization for Ms. Pac-Man

Link:

<https://ieeexplore.ieee.org/document/9654859>

3° ARTICLE:

Origin:

Reference of article 2

Article Title:

fairGhosts – Ant Colony Controlled Ghosts for Ms. Pac-Man

Link:

<https://ieeexplore.ieee.org/document/7744325>

Resume:

ChatGPT prompt: Summarize the following text in 300 words.

This paper presents fairGhosts, an improved ghost controller for the game Ms. Pac-Man based on **Ant Colony Optimization (ACO)**, a swarm intelligence algorithm inspired by how real ants find efficient paths to food. The goal is to create ghost agents that cooperate effectively to minimize Ms. Pac-Man's score while operating within the strict 40-millisecond decision limit imposed by the Ms. Pac-Man vs. Ghosts competition.

The proposed controller models the game maze as a graph and employs two types of artificial ants. ExplorerAnts investigate the maze and leave pheromone trails representing promising paths, while HunterAnts use this information to determine the best routes for the ghosts to pursue or evade Ms. Pac-Man, depending on whether they are in their normal or edible state. To encourage teamwork, ghosts reduce pheromone levels on their chosen paths so that other ghosts are more likely to select alternative routes, increasing the chances of trapping Ms. Pac-Man instead of following each other.

Compared with the original ACO framework, the authors simplified the algorithm, corrected the heuristic guiding ghost movement, removed unnecessary features, and

reduced the number of configurable parameters from ten to seven. They then optimized these parameters using a genetic algorithm, which evaluated many parameter combinations over repeated game simulations to identify those producing the strongest ghost performance.

To evaluate fairGhosts, the controller was tested against several leading Ms. Pac-Man and ghost controllers from the 2012 competition. Although a direct comparison with the original implementation was impossible due to changes in the simulator, fairGhosts consistently achieved competitive results. Its overall performance placed it within the upper third of all ghost controllers, with realistic potential to rank among the top ten.

The authors conclude that Ant Colony Optimization is an effective approach for developing cooperative, adaptive game AI. The fairGhosts controller demonstrates that swarm intelligence can produce strong real-time decision making in dynamic game environments while remaining computationally efficient and flexible enough for future improvements and applications beyond Ms. Pac-Man.

4° ARTICLE:

Origin:

Reference of article 1

Article Title:

*HearthBot: An Autonomous Agent Based on Fuzzy ART Adaptive Neural Networks for the Digital Collectible Card Game *HearthStone**

Link:

<https://ieeexplore.ieee.org/document/8015141>

Resume:

ChatGPT prompt: Summarize the following text in 300 words.

This paper presents HearthBot, an adaptive artificial intelligence (AI) agent designed to play the digital card game Hearthstone more effectively than traditional search-based methods. Hearthstone is a highly complex game with imperfect information, hidden cards, and an enormous number of possible game states, making it impractical to create optimal strategies manually or through exhaustive search. The authors estimate that the game contains approximately 2.85×10^{51} possible states, demonstrating the need for advanced machine learning techniques rather than conventional algorithms.

To address this challenge, the researchers model the game as a Partially Observable Markov Decision Process (POMDP) and represent game environments using key features such as player health, cards in hand, minions on the board, and battlefield strength.

Actions are encoded compactly based on card interactions and event categories, allowing the AI to efficiently process complex game situations.

The main contribution of the paper is an improved **Adaptive Resonance Associative Map (ARAM) neural network**. Instead of relying on traditional similarity measures that often overgeneralize different game situations, the authors introduce a normalized Manhattan distance metric to compare game states more accurately. This modification improves learning, prediction, and categorization while reducing data corruption and increasing the precision of decision making.

HearthBot follows a three-step process: it senses the current game state, predicts the reward of every possible action using the neural network, and selects the action with the highest predicted reward. During training, it learns by comparing simulated outcomes before and after each action, allowing it to refine its decision-making over time. The system was implemented using GPU acceleration to efficiently process millions of neural representations.

The researchers evaluated HearthBot against the **Monte Carlo Tree Search (MCTS)** agent provided by the Metastone simulator using multiple competitive Hearthstone decks. Experimental results showed that HearthBot consistently outperformed MCTS, achieving approximately 30% higher average win rates against familiar decks and around 21% higher win rates against previously unseen decks. The improved neural network also demonstrated strong generalization while maintaining efficient memory usage and scalable learning.

The authors conclude that combining an improved adaptive neural network with efficient state representation provides a powerful solution for decision making in complex, imperfect-information games. HearthBot demonstrates that adaptive learning can outperform traditional search methods while remaining capable of transferring learned knowledge to new situations, making the approach promising for future game AI research and other real-time decision-making applications.

3^a Week: Meeting 07/07/2026

5^o ARTICLE:

Search performed on the search engine:

Google Scholar

Search term:

Intelligent Adaptive Games

Selection Criteria:

Search term relevance

Article Title:

To Create Intelligent Adaptive Game Opponent by Using Monte-Carlo for Tree Search

Link:

<https://ieeexplore.ieee.org/abstract/document/5367088>

Resume:

ChatGPT prompt: Summarize the following text in 300 words.

This paper presents a method for creating intelligent and adaptive game opponents by combining **Monte Carlo Tree Search (MCTS)** with **Artificial Neural Networks (ANNs)**. The authors evaluate their approach using *Dead-End*, a two-dimensional predator-prey game in which two dog agents cooperate to capture a cat controlled by different player strategies. The objective is to improve non-player character (NPC) behavior by making opponents more adaptive and challenging while maintaining acceptable computational performance.

The proposed system first applies Monte Carlo Tree Search to determine the best move at each game state by simulating many possible future actions. Each legal move is explored through repeated simulations, and the move with the highest estimated success rate is selected. Experimental results show that increasing computation time generally improves MCTS performance up to a certain threshold, after which additional computation yields little or no benefit. However, because MCTS is computationally expensive, using it directly in real-time games can introduce unacceptable delays.

To overcome this limitation, the authors train Artificial Neural Networks using gameplay data generated by MCTS. Twenty-eight game-state features describing the positions and relationships between the cat, dogs, exits, and environment are used as ANN inputs. Separate neural networks are trained for different player strategies, allowing the dogs to

imitate the high-quality decisions previously discovered by MCTS while requiring significantly less computation during gameplay.

Cross-validation experiments demonstrate that each ANN performs best against the player strategy on which it was trained, although some networks generalize to similar strategies. The study concludes that neither technique alone is sufficient: MCTS is highly adaptable and effective against unknown strategies but computationally demanding, whereas ANNs are efficient and well suited for known strategies. Therefore, the authors recommend a hybrid approach in which MCTS generates training data and handles unfamiliar situations, while ANNs provide fast, adaptive decision-making during normal gameplay. This combination produces intelligent opponents that balance adaptability, efficiency, and real-time performance.

6° ARTICLE:

Search performed on the search engine:

Google Scholar

Search term:

Intelligent Adaptive Games

Selection Criteria:

Number of citations - 93

Article Title:

Computational intelligence in games

Link:

<https://www.cse.unr.edu/~bdbryant/papers/miikkulainen-2006-wcci.pdf>

Resume:

ChatGPT prompt: Summarize the following text in 300 words.

This paper explores how **neuroevolution (NE)**—the use of evolutionary algorithms to train artificial neural networks—can create adaptive, intelligent behaviors in video games. Unlike traditional scripted non-player characters (NPCs), neuroevolution enables game agents to learn, improve, and adapt through experience, producing more dynamic and unpredictable gameplay. The paper focuses on **NEAT (NeuroEvolution of Augmenting Topologies)** and its real-time extension, **rtNEAT**, which evolves neural network structures continuously during gameplay by replacing low-performing agents one at a time. This approach avoids abrupt behavior changes and allows agents to adapt seamlessly while the game is running.

To support adaptive AI in existing games, the authors integrate NEAT with the **TIELT** middleware and the game *Unreal Tournament*. The system enables neural-network-controlled bots to perceive their environment through sensors and evolve navigation behaviors, successfully learning to reach targets while avoiding obstacles. The experiments demonstrate that neuroevolution can reliably produce competent game behaviors, although improvements to middleware support, efficiency, and multi-agent functionality are recommended.

The paper also introduces **NERO (Neuro-Evolving Robotic Operatives)**, a machine learning game in which players train teams of AI-controlled agents instead of directly controlling them. Players design training scenarios by adjusting goals and rewards, allowing agents to evolve tactics such as seeking enemies, avoiding attacks, navigating mazes, and coordinating in combat. These experiments show that rtNEAT can rapidly generate sophisticated and often unexpected strategies while keeping gameplay interactive and engaging.

To make evolution more practical for commercial games, the authors propose **Knowledge-Based NEAT (KB-NEAT)**, which incorporates human knowledge into evolving neural networks. Existing behaviors, expressed as rules or finite-state machines, are converted into neural network structures and inserted into evolving agents. Experiments show that this prior knowledge significantly accelerates learning, remains adaptable when game objectives change, and gradually becomes integrated into the evolved behaviors. Overall, the paper concludes that combining neuroevolution with human guidance creates adaptive, efficient, and believable game AI, making it a promising approach for future intelligent game characters and interactive learning-based games.

7° ARTICLE:

Search performed on the search engine:

Google Scholar

Search term:

Intelligent Adaptive Games

Selection Criteria:

Number of citations - 56

Article Title:

Artificial Intelligence for Adaptive Computer Games.

Link:

<https://cdn.aaai.org/FLAIRS/2007/FLAIRS07-007.pdf>

Resume:

ChatGPT prompt: Summarize the following text in 300 words.

This paper examines how **Artificial Intelligence (AI)** can make computer games more adaptive, engaging, and believable. The authors argue that while graphics, sound, and animation have improved dramatically, game AI often remains simplistic, relying on scripted behaviors that can become repetitive and predictable. They propose that adaptive AI systems should learn from player interactions, adjust their behavior dynamically, and create richer gaming experiences. Beyond entertainment, such technologies have applications in education, military training, corporate simulations, and interactive storytelling.

The paper identifies several major challenges in game AI, including large decision spaces, knowledge engineering costs, limited authoring tools, unforeseen player behaviors, user-specific adaptation, replayability, and the need to satisfy educational or narrative objectives. The authors emphasize that **adaptive games** can both improve player experiences and reduce development effort by allowing systems to respond automatically to situations that designers did not anticipate.

To address these challenges, the paper presents three **Case-Based Reasoning (CBR)** approaches. The first focuses on **behavior modification for believable characters**. In this system, game characters monitor their own actions, detect when their behavior violates personality constraints, identify the cause of the problem, and modify their behavior during gameplay. This allows characters to adapt to changing situations while maintaining believable personalities.

The second approach applies **case-based planning to real-time strategy games**. Instead of requiring developers to hand-code strategies, the system learns from expert demonstrations. It stores examples of successful actions and later retrieves and adapts them to new situations, reducing programming effort while improving strategic behavior.

The third approach addresses **drama management in interactive stories**. By modeling player preferences from previous gameplay experiences, the system predicts which story developments a player is likely to enjoy and guides the narrative accordingly. This creates more personalized and engaging story experiences.

The paper concludes that adaptive AI, particularly through case-based reasoning, can transform games into more intelligent, immersive, and personalized experiences. Such advances may represent the next major evolution in the gaming industry, enabling developers to create games that learn, adapt, and respond dynamically to individual players.

8° ARTICLE:

Search performed on the search engine:

Google Scholar

Search term:

Intelligent Adaptive Games

Selection Criteria:

Publication Date - 2017

Article Title:

Artificial Intelligence for Adaptive Computer Games.

Link:

<https://ieeexplore.ieee.org/abstract/document/8324230>

Resume:

ChatGPT prompt: Summarize the following text in 300 words.

Overview of Game AI and Adaptability

This paper addresses the limitations of modern video games, which often exhibit high audiovisual realism but rely on non-adaptive Artificial Intelligence (AI). Non-adaptive methods (such as decision trees, finite state machines, and scripting) suffer from predictable patterns, allowing players to easily exploit opponent weaknesses. Academic research highlights that adaptive game AI can enable characters to learn player behaviors

and self-correct. However, developers are often hesitant to apply adaptive AI or online machine learning due to the risk of uncontrollable behaviors, extensive computational demands, and lengthy training trials.

Comparative Implementation in Unity

To demonstrate the transition from traditional to adaptive methodologies, the authors developed two games using the Unity engine:

- **"Danny's Nightmare" (Basic AI):** A 3D shooter game where a child fights waves of monstrous toys. The enemies (ZomBear, ZomBunny, and Zelephant) rely on a simple finite state machine. They lack a decision system for strategic evaluation and are limited to chasing and attacking the player directly upon sight, using pathfinding solely to bypass obstacles.
- **"Escape from Lab#98" (Adaptive AI):** An action-adventure stealth game where a character named Tony must escape a laboratory guarded by automated cameras, laser fences, and robots. It utilizes an elementary version of **Case-Based Reasoning (CBR)** driven by environment-aware "If" rules.

Key Differences and Conclusions

The major differentiation lies in tactical adjustment. Unlike the rigid enemies in "Danny's Nightmare," the robots in "Escape from Lab#98" dynamically alter their strategies. If they lose sight of the player, they dynamically re-route their patrolling scripts around the player's last known location.

Ultimately, the paper concludes that elementary adaptive AI enhances gameplay realism and notes that these practices can successfully migrate into real-world applications, including autonomous surveillance, intrusion detection, robotics, and predictive behavioral analytics.